GEOMETRY · QUADRILATERALS AND POLYGONS

Polygon & Quadrilateral Explorer

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Polygon & Quadrilateral Explorer** on Mr. K's Classroom: Geometry → Quadrilaterals and Polygons → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict

before you touch the tool

1. How many triangles do the diagonals from one vertex of a hexagon make?

4

2 Explore

use the tool and record what you see

Polygon angle sums

n	Name	Triangles	Interior sum	Each interior (regular)	Each exterior (regular)
3	triangle	**1**	**180°**	**60°**	**120°**
4	quadrilateral	**2**	**360°**	**90°**	**90°**
5	pentagon	**3**	**540°**	**108°**	**72°**
6	hexagon	**4**	**720°**	**120°**	**60°**
8	octagon	**6**	**1080°**	**135°**	**45°**
9	nonagon	**7**	**1260°**	**140°**	**40°**
12	dodecagon	**10**	**1800°**	**150°**	**30°**

Quadrilateral detective

Use the shape buttons, then check each property.

Shape	Parallel pairs	All sides ≅	4 right ∠s	Diagonals bisect	Diagonals ≅	Diagonals ⊥
Parallelogram	**2**	**no**	**no**	**yes**	**no**	**no**
Rectangle	**2**	**no**	**yes**	**yes**	**yes**	**no**
Rhombus	**2**	**yes**	**no**	**yes**	**no**	**yes**
Square	**2**	**yes**	**yes**	**yes**	**yes**	**yes**
Kite	**0**	**no**	**no**	**no**	**no**	**yes**
Isosceles trapezoid	**1**	**no**	**no**	**no**	**yes**	**no**

3 Write the rule

fill in each blank from what you observed

1. The interior angles of an n-gon add up to **(n − 2) · 180°** .
2. The exterior angles of any convex polygon add up to **360°** .
3. The diagonals of every parallelogram **bisect** each other.

4. A rectangle's diagonals are **congruent**. A rhombus's diagonals are **perpendicular**.

5. A square is both a **rectangle** and a **rhombus**.

4 Check no tool: show your work

1. Find the interior angle sum of a 10-gon.

   1440°

2. Each exterior angle of a regular polygon is 24°. How many sides?

   15

3. Find each interior angle of a regular 20-gon.

   162°

4. A quadrilateral's diagonals bisect each other and are perpendicular but not congruent. Name it.

   Rhombus

5 Reflect

Why do the exterior angles always add to 360°?

Walking around the polygon, you turn by each exterior angle and end up facing the way you started: one full turn.